Incidence Of Potentially Toxigenic Fungi And Aflatoxin In Maize (*Zea mays L.*) Infected By Maize Lethal Necrosis Disease In Bomet, Narok And Nakuru Counties Of Kenya

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ABSTRACT

Maize Lethal Necrosis Disease (MLND) is one of the most important biotic factors limiting maize productivity. MLND is believed to increase concomitant fungal infections that cause food poisoning in humans and livestock. This study aimed to establish the relationship between MLND and the associated fungal infections. Fifty four maize grain samples were obtained from three experimental sites in Bomet, Narok and Nakuru Counties during the maize growing seasons of 2019. A complete randomized experimental design with three replicates was applied using six hybrid maize varieties classified as resistant, tolerant and susceptible to MLND. The resistant varieties were KATEH16-02-R and DK777, tolerant (WE5135and WE5140), and the susceptible varieties (DUMA 43 and PH30G-20).Fungal colonies were isolated on Potato Dextrose Agar (PDA) medium amended with 25µg/l of streptomycin and tetracycline. Characterization of the fungal species was based on morphological and microscopic characterization. A total of 1861 fungal isolates were recovered from the samples, representing 82.25% *Fusarium verticilliodes*, 15.80% of Penicillium species 1.44% of *Aspergillus flavus*, 0.48% of *Aspergillus niger* and 0.2% of *Aspergillus parasiticus.* Fusarium was the most predominant fungi in all three study sites, with a mean relative density of 86%. Using Enzyme Linked Immunorbent Assay (ELIZA) method, Aflatoxin concentration levels of between 0.3 and 2.8 µg/kg were detected in 38.89 % of the samples. A significant positive correlation was established between MLND and aflatoxin levels ($r=0.7441$, $p<0.001$). These findings provide critical information on concomitant fungal species infection in maize infected with MLND and the potential risk of aflatoxins. Therefore there is a need to create awareness to the public about the detrimental effects of consuming MLND infected maize by both human and animals.

Key words: Maize Lethal Necrosis Disease, Fusarium, Pencillium, Aspergillus, Aflatoxin.

1. INTRODUCTION

Kenya's food security is significantly dependent on maize production since the crop is a key staple food to over 90% of the population, providing about 42% dietary energy intakes (Keya and Rubaihayo, 2013). Twenty five percent (25%) of maize is grown on large scale and seventy five percent (75%) is grown on small scale. Maize is used as human food, fodder and a source of raw materials for industrial products such as starch, maize syrup, cooking oil and biofuel. Despite the importance of the crop, the average yield of maize in many African countries and Kenya in particular remains as low as 1-15t/ha-which is only about a third of the world average (Smale *et al.*, 2011). A significant portion of yield loss is attributed to the effects of prevalent diseases including Maize Lethal Necrosis Disease (MLND) among other viral and fungal diseases, pests (Fall army worms and stem borers) and toxins such as aflatoxins (AGRA, 2014).

Maize lethal necrosis disease which is caused by a co-infection of maize with maize chlorotic mottle virus, maize Sugar cane mosaic mosaic virus, dwarf mosaic virus or wheat streak mosaic virus and transmitted by